HMC ONE
AI STRATEGY. BUILT TO SCALE.

CLIENT CASE STUDY · CONSUMER DURABLES MANUFACTURING · ROBOTICS

AG-3X THREE-AXIS CARTESIAN GANTRY · INTERACTIVE 3D MODEL

Spec Sheet to Spinning Model

A GenAI-built, browser-based model of the AG-3X gantry turned a spec sheet into a machine customers can spin, explode and jog, on any device. Engineers supplied the facts; Claude wrote the 3D code.

~~5 wk~~ → **3 days** spec sheet to interactive model, vs animation

3 days
Spec sheet to interactive model
▼ vs ~5 weeks for animation

80 KB
Size of the complete model
one file, opens on a phone

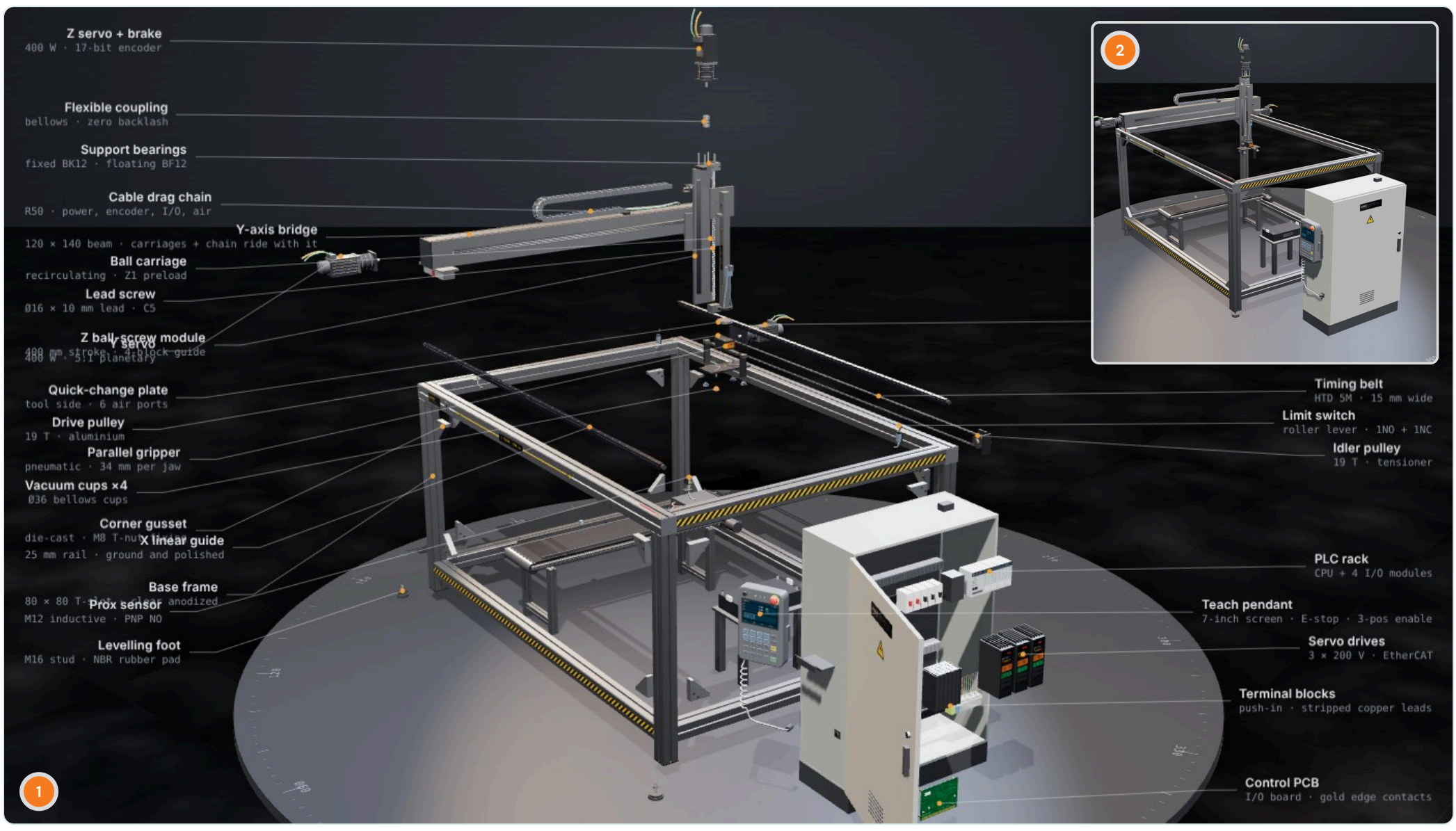
26 parts
Components labelled with specs
▲ across 5 build stages

18 /month
Customer demos per month
▲ vs 5 with static decks

Rings: 3 days against ~5 weeks (inner ring); one single file, no install; 26 parts split by build stage (drive train in orange); 18 demos a month with the earlier 5 inside. Figures indicative; validated metrics available on request.

01 The model, as delivered

Stills rendered from the actual single-file WebGL model



1 **Exploded view** at 100%: each sub-assembly lifts apart, with every part named and specced from the BOM

2 **Assembled** on the turntable: frame, Y bridge, Z module, cabinet and teach pendant

3 **Controls in the file**: jog X, Y, Z and the gripper, run the demo cycle, drag to turn, pinch to zoom

02 26 parts, five build stages

The model follows how the machine is built, so it doubles as a teaching tool

01 Frame

Base frame
80 × 80 T-slot · clear anodized

Corner gusset
die-cast · M8 T-nut fixing

Levelling foot
M16 stud · NBR rubber pad

02 Motion axes

Y-axis bridge
120 × 140 beam

X linear guide
25 mm rail · ground and polished

Ball carriage
recirculating · Z1 preload

Cable drag chain
R50 · power, encoder, I/O, air

Z ball-screw module
400 mm stroke · 4-block guide

Prox sensor
M12 inductive · PNP NO

Limit switch
roller lever · 1NO + 1NC

03 Drive train

X servo + gearbox
750 W · 10:1 planetary

Y servo
400 W · 5:1 planetary

Z servo + brake
400 W · 17-bit encoder

Timing belt
HTD 5M · 15 mm wide

Drive pulley
19 T · aluminium

Idler pulley
19 T · tensioner

Flexible coupling
bellows · zero backlash

Lead screw
Ø16 × 10 mm lead · C5

Support bearings
fixed BK12 · floating BF12

04 End effector

Quick-change plate
tool side · 6 air ports

Parallel gripper
pneumatic · 34 mm per jaw

Vacuum cups ×4
Ø36 bellows cups

05 Controls

PLC rack
CPU + 4 I/O modules

Servo drives
3 × 200 V · EtherCAT

Terminal blocks
push-in

Control PCB
I/O board

STROKE X × Y × Z	PAYLOAD	REPEATABILITY	X DRIVE	Y DRIVE
1900 × 1200 × 400 mm	12 kg	±0.02 mm	750 W servo, 10:1	400 W servo, 5:1
Z DRIVE	**LINEAR GUIDES**	**FRAME**	**DRIVES BUS**	**END EFFECTOR**
400 W, brake	25 mm rails	80 × 80 T-slot	EtherCAT	Gripper + 4 vacuum cups

AG-3X at a glance · specifications as labelled in the model

Every label taken from the BOM: part name plus its key spec

03 Four pain points, four fixes

How the machine gets explained · before → after

BEFORE
CAD stays in CAD
Design files needed licensed software and powerful laptops. Customers and sales could not open them.

SINGLE HTML FILE

AFTER
Any browser
One ~80 KB file in any browser, including phones, shared as a link.

BEFORE
Static renders, no motion
Slides showed the gantry, not how X, Y and Z move or what the gripper does.

THREE.JS · WEBGL

AFTER
Motion on demand
Jog X, Y, Z and the gripper; watch the demo cycle; explode 26 labelled parts.

BEFORE
Animation was slow
An outsourced walkthrough took weeks, and every design change meant paying for a re-render.

CLAUDE WRITES THE CODE

AFTER
Changes in hours
Update the code with Claude in hours; labels kept in step with the BOM.

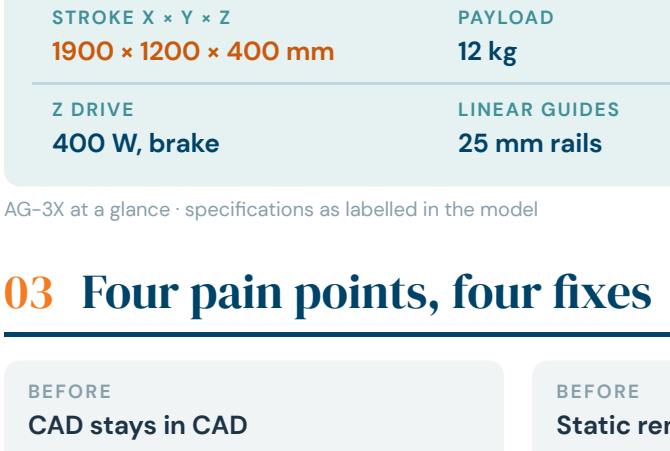
BEFORE
Training waited for steel
Operators and service teams could only learn the machine once a physical unit existed.

BUILD STAGES

AFTER
Learn before delivery
Walk through the five build stages before the machine arrives.

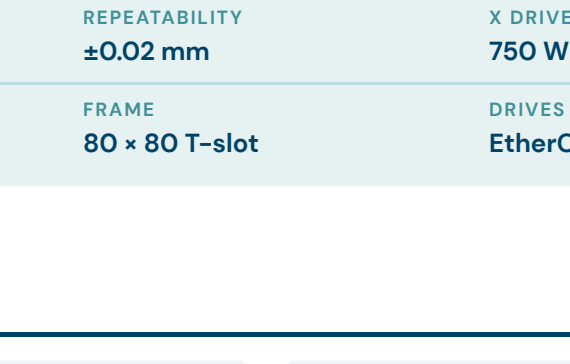
04 Hours per visual asset

Before (open dot) and after (filled) · indicative



05 Who opens the model

Sessions by audience, first 8 weeks · indicative



135 sessions in the first 8 weeks; the design team is only 24 of them

06 How HMC ran it

2 days — Discover
Sales, design, service and training asked what they need to see; spec sheet, BOM and images collected.

1 day — Design
Storyboard of five build stages; label list from the BOM; pendant-style controls.

3 days — Build
Claude generated the single-file Three.js scene; engineers reviewed each round.

4 weeks — Adopt
Sales demos and shared links, design reviews, pre-delivery training.

∞ ongoing — Scale
Model template, prompt pack and review checklist for future machines.

What transfers to any machine builder

01 — A link beats a render
Customers engage longer with a machine they can move. A shareable file keeps the conversation going after the meeting.

02 — Let the BOM drive the model
Every label came from engineering data. GenAI wrote the code; engineers kept it accurate.

03 — Build once, template forever
The first model took the learning. The next machine starts from a working template and a tested prompt pack.

Built with **Claude** · **Three.js** (WebGL) · **Spec sheet & BOM** · **Single HTML file** | Prepared by Havish M Consulting · Figures indicative; validated metrics available on request.

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